WSABI Summer Research Lab **Summer 2026**

Lab 19: Kernel Methods

Instructor: Jonathan Pipping-Gamón *Authors: JP, RB*

19.1 Soccer Shot Density and Expected Goals

In this lab, you will use kernel methods to study Premier League shots from the 2024–2025 season. The data file is `19_shots.csv`. Each row represents one shot from the `xg-plus` tracking data project.

The response variable is `goal`, which equals 1 if the shot became a goal and 0 otherwise. The location columns `shot_x` and `shot_y` place every shot on a common attacking pitch: `shot_x` runs across the pitch, while `shot_y` runs toward the attacking goal at the top. The file also contains features derived from tracking data, such as:

- distance and angle to goal;
- ball height and ball speed;
- goalkeeper distance and angle;
- open goal share;
- nearest defender and teammate distances.

The rows are split by match date into `train`, `validation`, and `test`. Use the training split to fit each model, the validation split to tune bandwidth choices, and the test split only for final evaluation.

19.1.1 Task 1: Explore Shot Locations

1. Load `19_shots.csv`. Report the number of shots and goals in each split.
2. Make a pitch plot of all shot locations. Use a different color or shape to identify goals.
3. Summarize the distributions of distance to goal, absolute angle to goal, open goal share, and nearest defender distance. Which features appear most related to goal probability?

19.1.2 Task 2: Kernel Density Estimation

Use kernel density estimation to visualize where shots are taken.

1. Create at least three KDE shot density plots using different bandwidths.
2. Explain how the bandwidth changes the estimated shot density surface.
3. Identify one region where the small bandwidth estimate appears noisy and one region where the large bandwidth estimate appears too smooth.

19.1.3 Task 3: Location Only Kernel Regression

Fit a location only expected goals model using the Nadaraya–Watson estimator:

$$\widehat{p}(\boldsymbol{x}) = \frac{\sum_{i \in \text{train}} K_h(\boldsymbol{x} - \boldsymbol{x}_i) y_i}{\sum_{i \in \text{train}} K_h(\boldsymbol{x} - \boldsymbol{x}_i)}.$$

Here $\boldsymbol{x}_i = (\texttt{shot\_x}_i, \texttt{shot\_y}_i)$.

1. Write a function that computes Gaussian kernel predictions for new shot locations.
2. Try a grid of location bandwidths. For each bandwidth, fit on the training split and compute validation log loss.
3. Choose the bandwidth with the lowest validation log loss.
4. Refit the model using the training and validation splits together, then evaluate test log loss.
5. Create a pitch heatmap of the location only xG surface and interpret it.

19.1.4 Task 4: Add Tracking Feature Similarity

Now define similarity using more than shot location. Use a feature set that includes location or features derived from location, along with several features derived from tracking data. A reasonable starting set is:

`distance_to_goal`, `abs_angle_to_goal`, `ball_height`, $\log(1 + \texttt{ball\_speed})$,
`goalkeeper_distance`, `goalkeeper_abs_angle`, `open_goal_share`, `nearest_defender_distance`.

1. Standardize the features using means and standard deviations computed from the training split. Apply the same transformation to the validation and test splits.
2. Tune a Gaussian kernel bandwidth in the standardized feature space using validation log loss.
3. Refit the model using the training and validation splits together, then evaluate test log loss.
4. Compare the location only and richer feature models using test log loss.
5. Make a scatter plot comparing the two models' test set predictions. Which shots move up or down when tracking features are included?